

CAT 304W - Group Innovation Project and Study for Sustainability

**Semester 1
Academic
Session 2025/26**



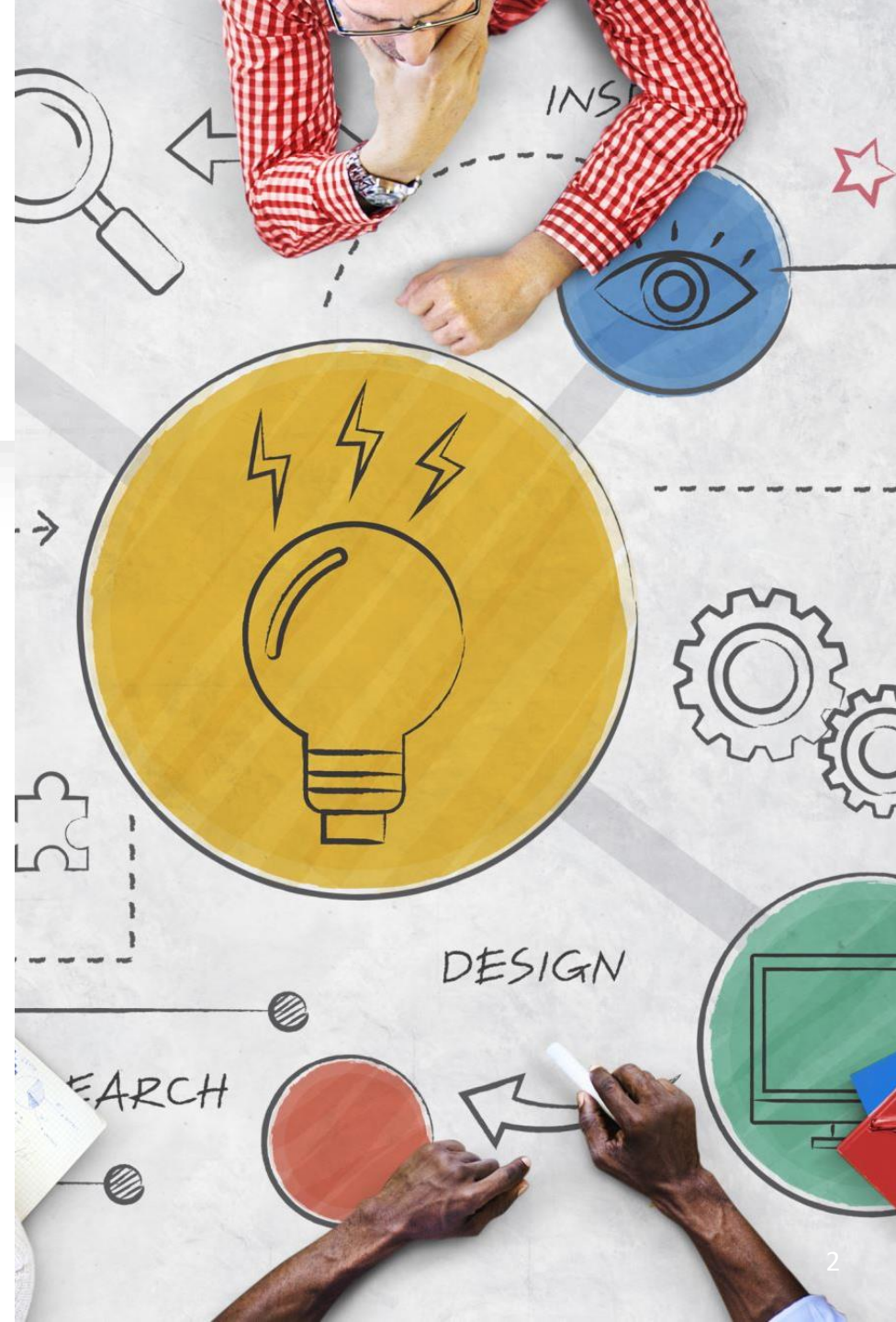
**UNIVERSITI
SAINS
MALAYSIA**

School of Computer Sciences



Course Introduction

- GROUP *innovation project*
- & Study for Sustainability



Course Coordinator:

- Mohd Azam Osman (MA) (Assoc. Prof.)
- Dr. Wong Khang Siang

Course Supervisors:

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Objective

To expose students to the basic skills in conducting professional software development projects and technical study in a group that is infused with innovation and sustainable elements

Course Synopsis

The course allows students to acquire training in developing professional software using a systematic innovation based on TRIZ and sustainability through a group project. This course also trains students to develop a system, write a technical paper and software documentation, perform effective presentation and demonstration of a project.



Group *innovation project* & Study for Sustainability

It's a group project

To propose and develop a new or innovative computer solution that resolves the problem

The proposed solution is not only addressing the immediate problem, but it can also provide a long-term solution.

Timeline

	Week	Date
Form and submit a group members	1	13-Oct-2025
Group meeting with supervisor	2	Date & Time - arrange with supervisor - Discuss on the proposed idea
Extended proposal submission	4	29-31 Oct-2025
Group meeting with supervisor	6	Date & Time - arrange with supervisor - Discuss on the implementation of the project - Enhancement/refining the proposed project proposal
Group meeting with supervisor	10	Date & Time - arrange with supervisor - Update the project implementation - Final report preparation
Submit a draft of Final report	13	Submit a draft of final report to the supervisor - Discuss the improvement of a final report.
Final report submission	14	5-Jan-2026
Project presentation & demo	15	13 - 14 Jan 2026

Note: During the group meeting, the group can discuss any issues related to the project.

Assessments

100% Coursework

Coursework Evaluation	Report (37)	25%
	Project (32)	50%
	Presentation (53)	10%
	Involvement in planning (41)	15%

Evaluation



Evaluation Components		
1. Group Innovation Project	%	
Extended proposal	10%	85%
Technical Report	15%	
System	50%	
Presentation	10%	
2. Participation + group dynamics	15%	15%
Total	100%	100%

****Participation + group dynamics -**

- 10 % Peer review
- 5 % Supervisor

Evaluation

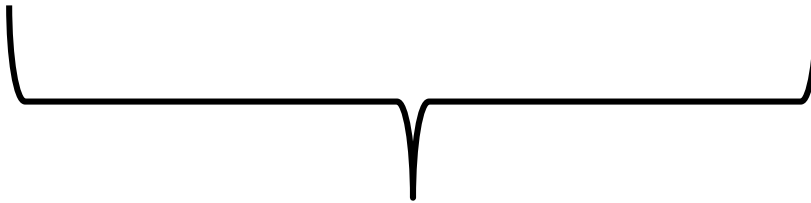


Group

+



Individual



Peer Review + Supervisor + Appointed Panels/examiners



Rubric



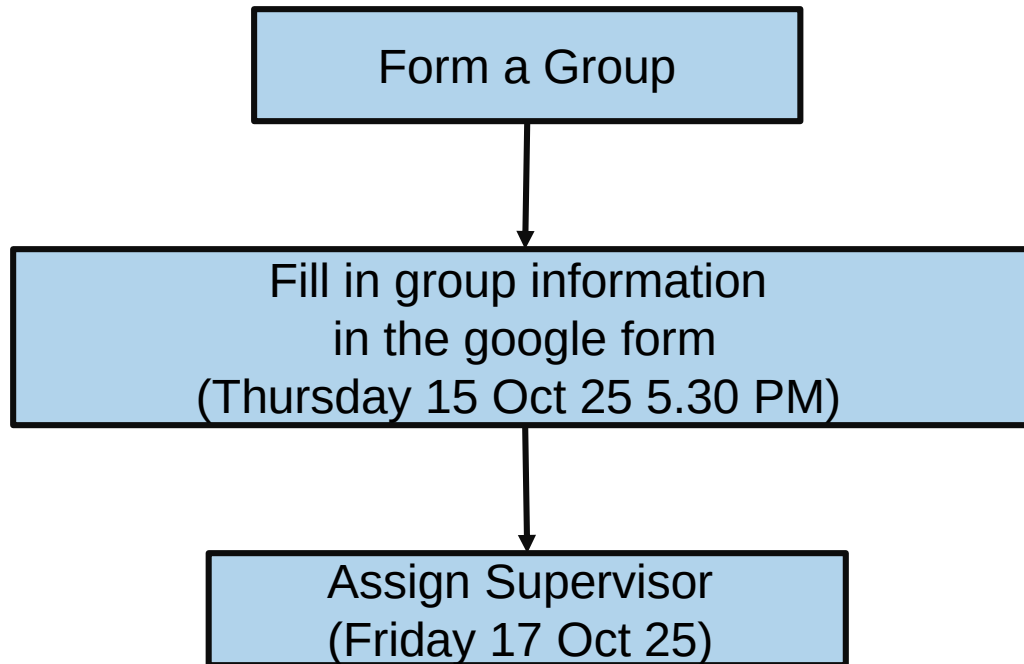


Do not Panic or Stress



4 students per group

+ - 1 student with proper justification



Idea to Project



Based on SDGs,
MyStie 10-10.
Including Malaysia's
Madani concepts





SUSTAINABLE DEVELOPMENT GOALS

17 GOALS TO TRANSFORM OUR WORLD



<https://sdgs.un.org/goals>

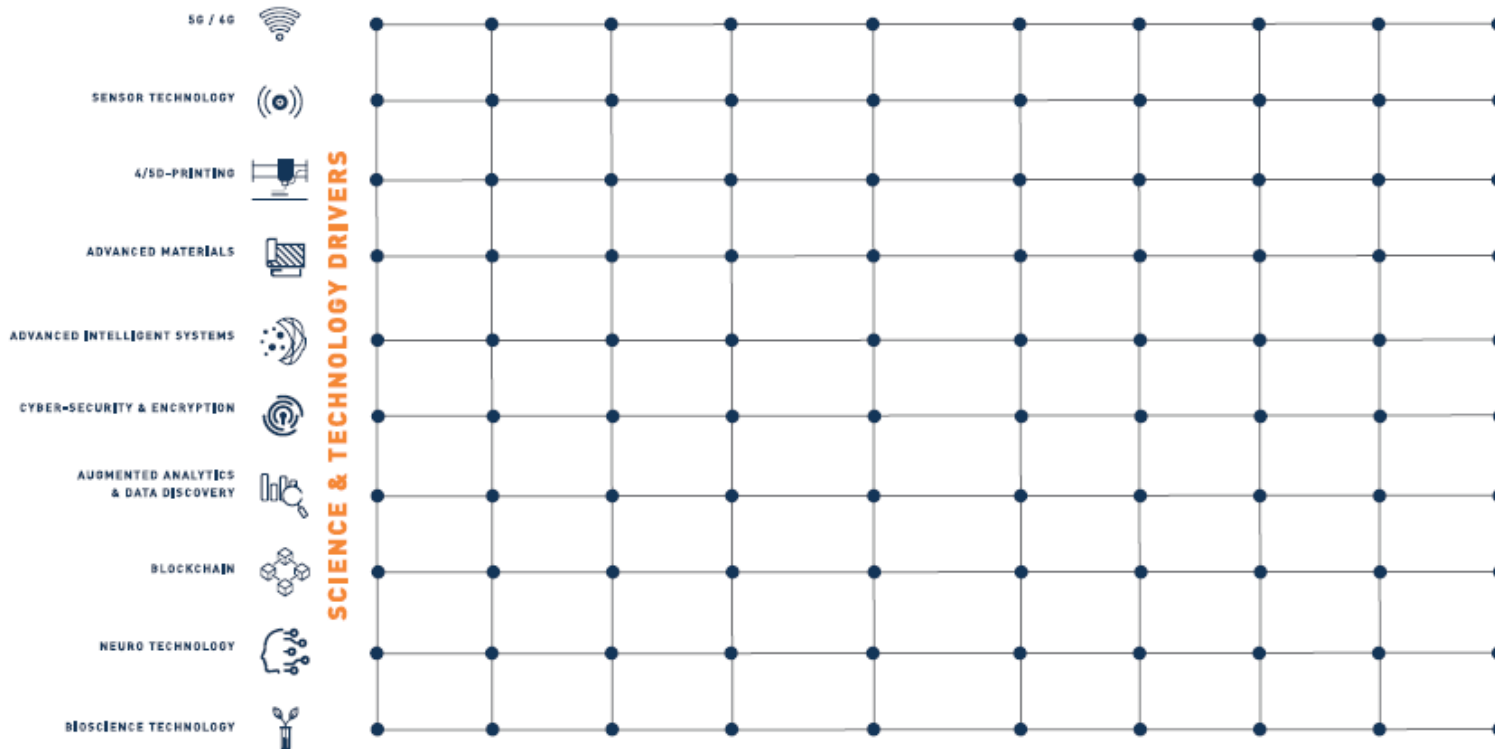


$\frac{10}{10}$ | MySTIE FRAMEWORK

Trailblazing the Way for Prosperity,
Societal Well-Being & Global Competitiveness

<https://issuu.com/asmpub/docs/10-10-mystie>

MALAYSIAN SOCIO-ECONOMIC DRIVERS



Each Science & Technology Driver should explore core technologies & applications for the 10 Malaysian Socio-economic Drivers

Driving Fundamental & Translational Research

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Each Malaysian Socio-economic Driver should explore how the 10 Science & Technology Drivers will value-add and enhance their global competitiveness

What is Innovation?

- The process of turning an idea or invention into a good or service that creates value or is worth paying for.
- To be considered an innovation, an idea must be replicated economically and meet a specific need
- Innovation involves the deliberate application of information, imagination, and initiative to derive greater or different values from resources. It encompasses all the processes that generate new ideas and turn them into useful products.

What is Innovation?

a new idea or method, or the use of new ideas and methods

Source: **Cambridge Dictionary**

How to Be More Innovative

"In aiming for innovation, surround yourself with those that think differently than you. Doing so will allow you more perspectives than what you would be capable of. The old adage of "Great minds may think alike" but great inventions come from multiple great minds."

- Siam Luu

Creative and Innovative Solution



Ordinary Solution

Example: Airline Industry



VS



Innovation

- Marketing – Everyone can Fly
- Purchasing Ticket – online & offer
- Flight Convenient – passenger choice



Purchase Ticket at counter



Online Ticket

Anything else you can share???

Example: Food Industry



Innovation



Food order => Food Panda



Education while Covid-19 around



Online Education Software/Application



- **Getting Idea**



Study the current practice & challenges/issues



Define and understand the problem domain



Study the existing systems/solutions



Identify opportunity - the solution / improvement



Techniques/methods



Technology to be deployed – h/w & software



Feasible & Practical (cost, ease of use, implementation readiness)

Possible project



WEB APPLICATION



WEBSITE



VS



AI based project

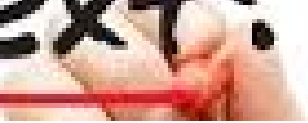


Information System

Cyber/Data Security

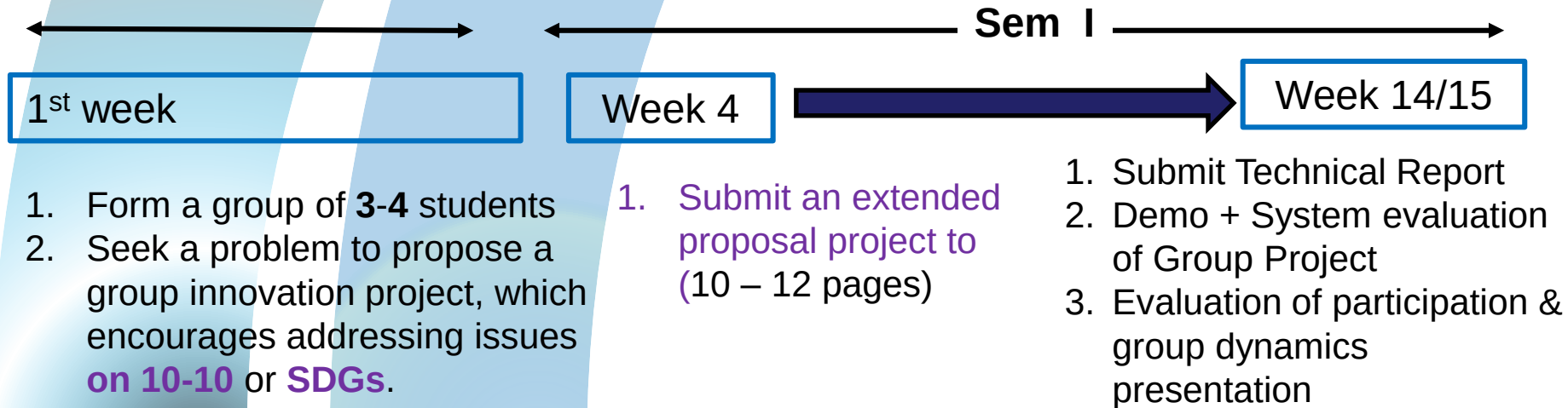


What's Next?



1. **Form a team (select a leader)**
2. **Discuss the proposed project with your teammate and then proceed to discuss it with yourself**
3. **Prepare and submit Extended Proposal**
4. **Implement the project**
5. **Prepare Final Report**
6. **Final group Presentation (Study week)**

(CAT 304): Group Innovation Project Implementation



- Students are encouraged to suggest their own project. Students can also continue the project from CSE 241

****Participation + group dynamics**

- 10 % Peer review
- 5 % Supervisor

10 %

+

90%

Evaluation Components		
1. Group Innovation Project	%	
Extended proposal	10%	85%
Technical Report	15%	
System	50%	
Presentation	10%	
2. Participation + group dynamics	15%	15%
Total	100%	100%

A hand in a blue suit sleeve holds a white, torn-edge paper. The paper has the text 'Preparing Extended Proposal' in black. Below the paper is a red box. The background is a blurred blue suit.

Preparing Extended Proposal

Extended Proposal

CAT304 Extended Proposal

Academic Session: 2025/2026

Front page

CAT304

**Group Innovation Project and Study for Sustainability
Initial Project Proposal**

[Project Title]

[List of team members Name]
[list of e-mail address]

[Supervisor Name]

School of Computer Sciences, Universiti Sains Malaysia
11800 USM, Penang, Malaysia

[Group No.]

Let's see

*Refer to the given guidelines in
eLearning*

Final Report



**SCHOOL OF COMPUTER SCIENCES
UNIVERSITI SAINS MALAYSIA**

CAT304

Group Innovation Project and Study for Sustainability

**Final Report
E-venture**

Name	Matric No.	E-mail address
Cheng Chea Hao		
Lim Ting Juin		
Choong Yew Chung		
Choong Jun Yang		

**School of Computer Sciences, Universiti Sains Malaysia
11800 USM, Penang, Malaysia**

Group 9 - F4

Supervisor: Assoc. Prof. Mohd Azam Osman



Report Content

- i. Declaration - claim of the originality your work
- ii. Acknowledgements
- iii. Abstract – (in Bahasa & English)
- iv. Table of Contents (including References & Appendix)
- v. List of Tables
- vi. List of Figures
- vii. List of Abbreviations and Symbols

1.0 Introduction

2.0 Background Study and Related work

3.0 System Requirements & Analysis

4.0 System Design

5.0 System Implementation

6.0 Testing & Evaluation

7.0 Conclusion and Future work

Final Report

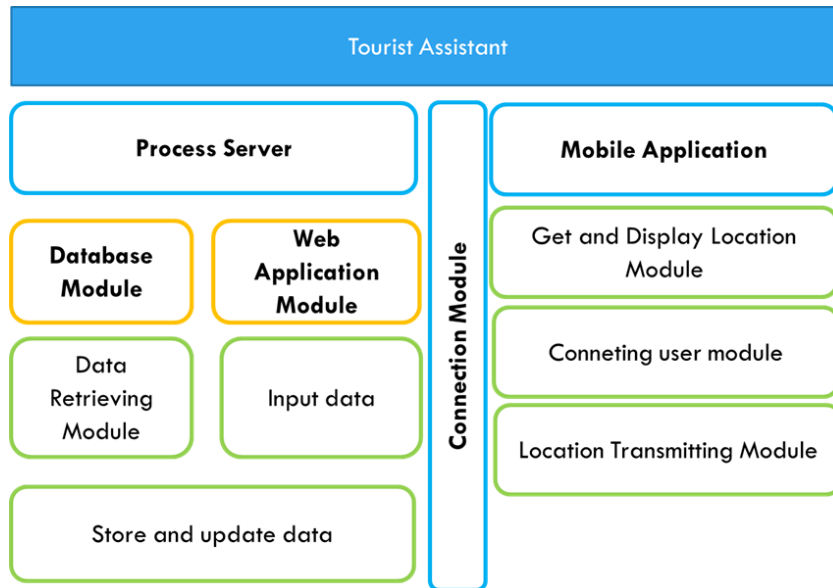


1.0 Introduction

- Background –general background of the project environment
- Problem statement – state the problems within the project environment
- Proposed solution / project
 - Should include overall modules of the proposed solution / project
- Project objectives
- Motivation
- Benefit or impact of the proposed solution
- Uniqueness of proposed solution (innovation or novelty)
- Contribution
 - List of tasks/works for each individual
 - List of tasks/works done together
 - Based on work breakdown structure and the overall module of the proposed solution / project – highlight in colour which modules belong to each individual
- Organization of the report

1.0 Introduction

- Contribution
 - List of tasks/works for everyone
 - List of tasks/works done together
 - Based on work breakdown structure and the overall module of the proposed solution / project – highlight in colour which modules belong to each individual
- Organization of the report



1.0 Introduction

- **Background –general background of the project environment**

Discuss on current trend or current practice which related to your proposed project . Statements/facts should be supported by references such as magazines, news papers or research references (journal or proceeding).



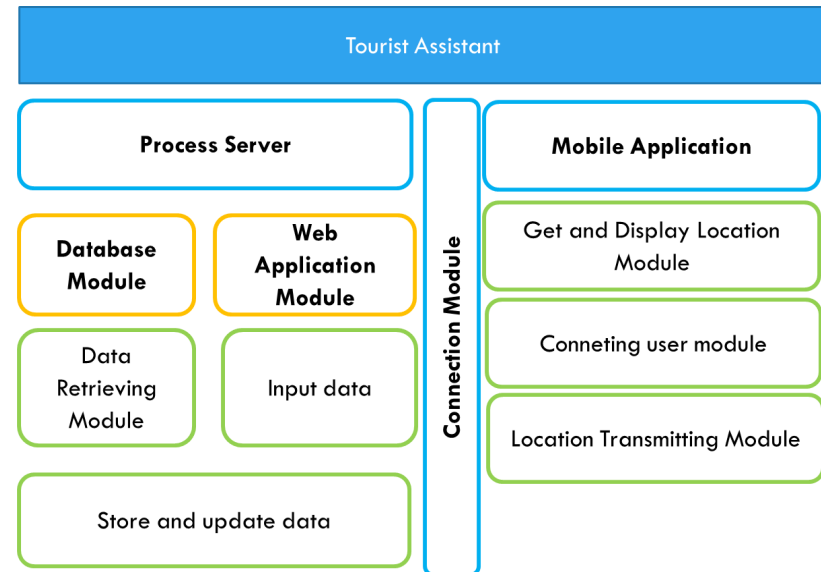
- **Problem statement (1 paragraph)**
state the problems within the project environment . Usually, problems statements should be supported by references.

1.0 Introduction

- **Proposed solution / project**

Tourist assistant apps is a **mobile application** for travelers, travelers' family members and friends to **assist** them when they going for a **vacation**. Tourist assistant consists of **3 main components** that **is mobile application module, connection module and process server**. The diagram x.x shows the overall modules/components of the Tourist assistant apps. Tourist assistant apps will provide the following main features:

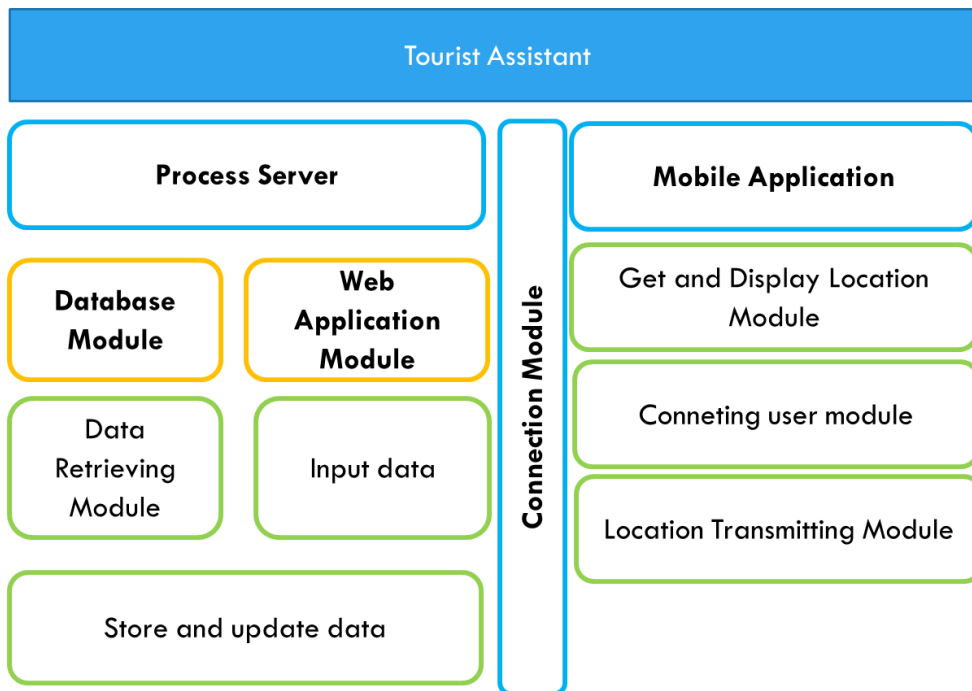
- **Recommendation**
 - To **provide the attractions information** to the user. User are able to check for **public transport** available to the destination and explore the city in a better way.
- **Sharing location**
 - To **show friends nearby location on a map**. This feature enables users that travel in a group to know where are their friends' location when each of them moving by themselves. User will have fully control on the location sharing system, so that their privacy risk and control is on their hand.
- **Sharing moment**
 - To enable the user to **share the excited moment** and joy with other user. Sharing delicious food and beautiful picture within each other.



1.0 Introduction

- **Team Member Contribution**

- List of tasks/works for everyone
- List of tasks/works done together
- Based on the overall module of the proposed solution / project
 - highlight in colour which modules belong to each individual



**Important!!! –
Evaluation will be
based on
group & individual
contribution**

EXAMPLE

1.0 Introduction

Problem Statement

According to worldwide statistics, shown in research conducted by the Health and Safety Executive (HSE) in the United Kingdom, 526,000 workers have suffered from work-related stress, depression, or anxiety (new or long-standing) in recent years. Furthermore, an estimated 12.5 million working days were lost due to work-related stress, depression, or anxiety [4].

Proposed project/solution

Given all the problems that have been stated in the previous section, we propose The Eye 2A - Affectivity Analysis for Working Spaces. The Eye 2A is an application that utilizes computer vision and image processing algorithms to identify emotions of target individuals. The Eyes 2A can assist the right authority in analysing and understanding the emotions of employees, which can reduce conflicts between employees and improve productivity within the organization. This application consists of three main modules: the user information module, the cognitive computation module, and the data analytics module. Figure 5 shows the overall module of the applications.

1.0 Introduction

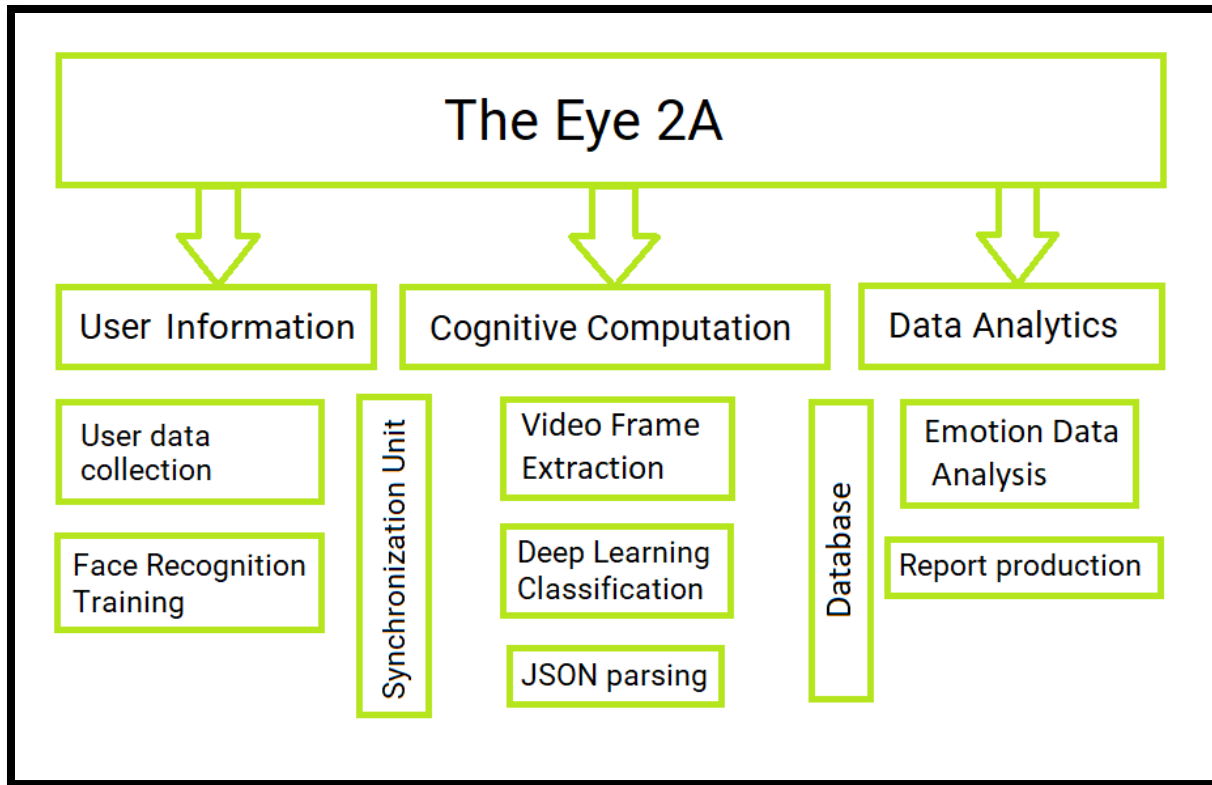
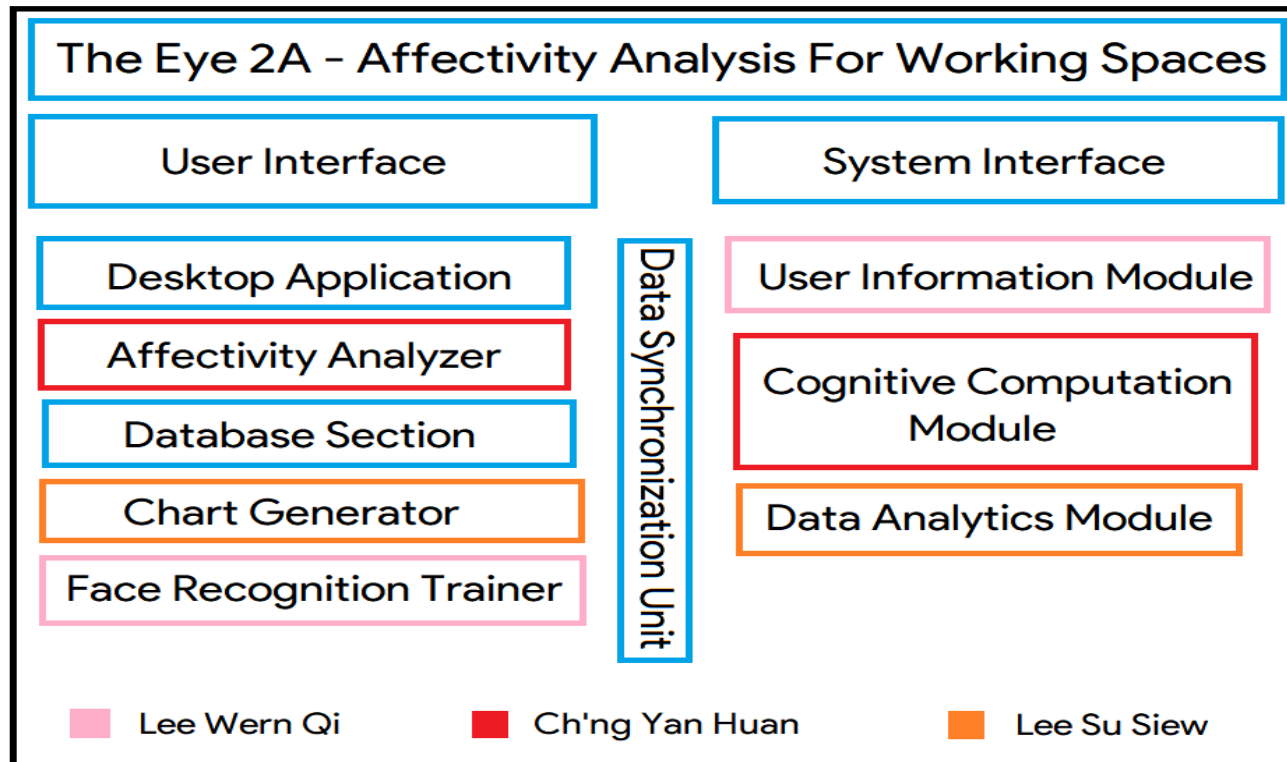


Figure 5: Overall Module of The Eye 2A

1.0 Introduction

- **Team Member Contribution**

- List of tasks/works for everyone
- List of tasks/works done together
- Based on the overall module of the proposed solution / project
 - highlight in colour which modules belong to each individual



2.0 Background Study & Related Work

- Existing System / Project / Application (similar and related your proposed solution)
- Features Comparison of the existing System / Project / Application
- Existing Technique/Algorithm/Method



3.0 System Requirements and Analysis

- Status of project development (new or continuation/enhancement)
- Scope of the proposed solution
- System Capabilities & Limitations
- Project Management
 - Work Breakdown Structure
 - Gantt Chart
 - SWOT Analysis
- Development Methodology – **OO/Waterfall/ Spiral/Agile**



3.0 System Requirements and Analysis

Analysis of Proposed Solution/Project

- Identify users' requirements
 - There are many ways to seek users' requirements. State and justify the chosen ways.
- UML diagram
 - Use Case Diagram
 - Use Case Description
 - Activity Diagram
 - Sequence Diagram (Optional)
 - State Chart Diagram (Optional)
 - Flow chart Diagram (optional)



3.0 System Requirements and Analysis

- **Technology Deployed**
 - Hardware Specification
 - Software Specification - including OS system
 - Programming Languages / Tools – (with justification)
 - Chosen Algorithm/method/technique & justification (if relevant)



4.0 System Design

- System architecture (diagram and explanation)
- System components / modules
- Database design & Class Diagram – if required
- Interface design – User Interface Hierarchy
- Input & Output Design



5.0 System Implementation

- Module implementation
 - Based on Individual Contribution , each of team member of the project should describe on how the project is implemented
 - E.g., how do you implement each module?
 - May be some code explanation
 - Algorithm (flow chart or Pseudo code), story board,
..
- Describe system integration. How do you integrate overall modules?



Testing / Evaluation and Conclusion & Future Work

6.0 Testing / Evaluation

System Testing (unit testing, Integration Testing, System Testing)

Test Case

System Evaluation

7.0 Conclusion & Future Work



CAT304W EXPECTATION

Presentation

Exterior

User interface, handheld devices adaptability, multi-form of input & output , ...

Interior

Database, Algorithms, system design & architecture (cloud computing, ..)

Performance

Coding techniques, multi-core, GPU, Cloud, etc.

Technology Exploration / Precision

Sensors, Arduino, Galileo Intel, Xbox Kinect tech. ultra sonic sensors and more

Documentation

A hand is holding a piece of brown cardboard. On the cardboard, the words "Time for" are written in white, and "Questions" is written in blue. The text is slightly tilted to the right.

**Time for
Questions**

any questions?